

Chapter 5

Large and Fast: Exploiting Memory Hierarchy

Principle of Locality

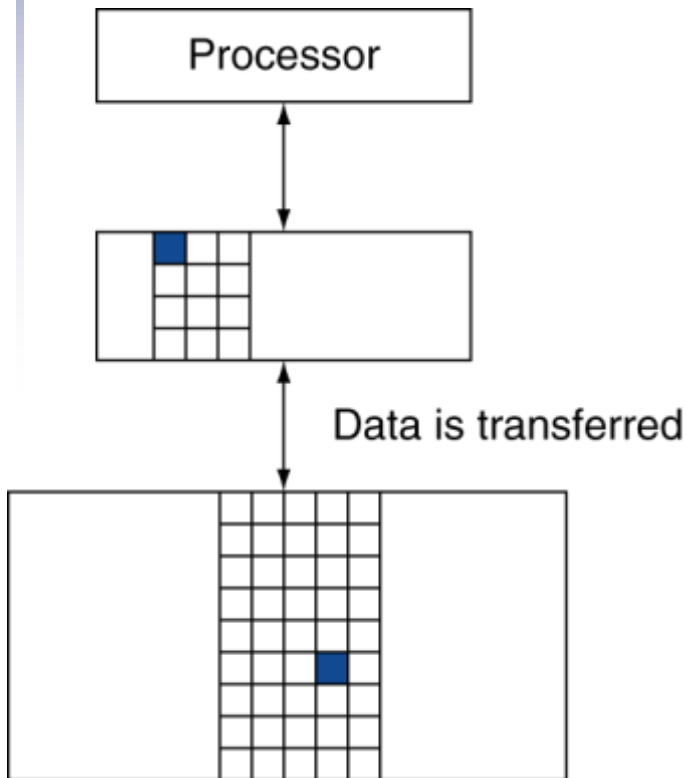
- Programs access a small proportion of their address space at any time
- Temporal locality
 - Items accessed recently are likely to be accessed again soon
 - e.g., instructions in a loop, induction variables
- Spatial locality
 - Items near those accessed recently are likely to be accessed soon
 - E.g., sequential instruction access, array data

Taking Advantage of Locality

- Memory hierarchy
- Store everything on disk
- Copy recently accessed (and nearby) items from disk to smaller DRAM memory
 - Main memory
- Copy more recently accessed (and nearby) items from DRAM to smaller SRAM memory
 - Cache memory attached to CPU

Memory Hierarchy Levels

- Block (aka line): unit of copying
 - May be multiple words
- If accessed data is present in upper level
 - Hit: access satisfied by upper level
 - Hit ratio: hits/accesses
- If accessed data is absent
 - Miss: block copied from lower level
 - Time taken: miss penalty
 - Miss ratio: misses/accesses
 $= 1 - \text{hit ratio}$
 - Then accessed data supplied from upper level



Memory Technology

- Static RAM (SRAM)
 - 0.5ns – 2.5ns, \$2000 – \$5000 per GB
- Dynamic RAM (DRAM)
 - 50ns – 70ns, \$20 – \$75 per GB
- Magnetic disk
 - 5ms – 20ms, \$0.20 – \$2 per GB
- Ideal memory
 - Access time of SRAM
 - Capacity and cost/GB of disk

Cache Memory

- Cache memory
 - The level of the memory hierarchy closest to the CPU
- Given accesses $X_1, \dots, X_{n-1}, X_n$

X_4
X_1
X_{n-2}
X_{n-1}
X_2
X_3

a. Before the reference to X_n

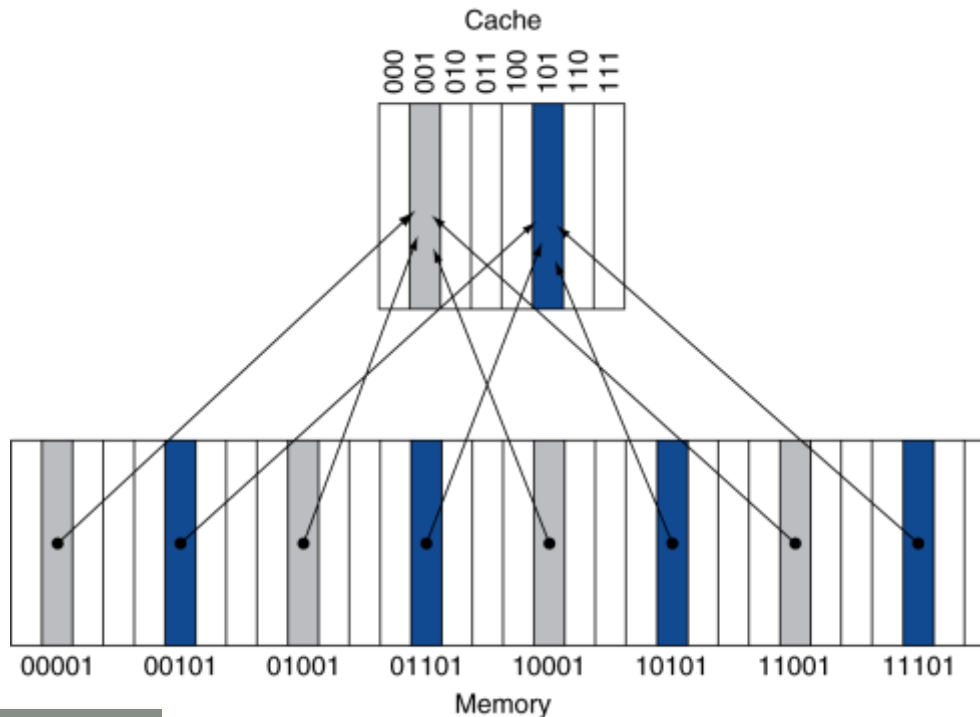
X_4
X_1
X_{n-2}
X_{n-1}
X_2
X_n
X_3

b. After the reference to X_n

- How do we know if the data is present?
- Where do we look?

Direct Mapped Cache

- Location determined by address
- Direct mapped: only one choice
 - (Block address) modulo (#Blocks in cache)



- #Blocks is a power of 2
- Use low-order address bits

Tags and Valid Bits

- How do we know which particular block is stored in a cache location?
 - Store block address as well as the data
 - Actually, only need the high-order bits
 - Called the tag
- What if there is no data in a location?
 - Valid bit: 1 = present, 0 = not present
 - Initially 0

Cache Example

- 8-blocks, 1 word/block, direct mapped
- Initial state

Index	V	Tag	Data
000	N		
001	N		
010	N		
011	N		
100	N		
101	N		
110	N		
111	N		

Cache Example

Word addr	Binary addr	Hit/miss	Cache block
22	10 110	Miss	110

Index	V	Tag	Data
000	N		
001	N		
010	N		
011	N		
100	N		
101	N		
110	Y	10	Mem[10110]
111	N		

Cache Example

Word addr	Binary addr	Hit/miss	Cache block
26	11 010	Miss	010

Index	V	Tag	Data
000	N		
001	N		
010	Y	11	Mem[11010]
011	N		
100	N		
101	N		
110	Y	10	Mem[10110]
111	N		

Cache Example

Word addr	Binary addr	Hit/miss	Cache block
22	10 110	Hit	110
26	11 010	Hit	010

Index	V	Tag	Data
000	N		
001	N		
010	Y	11	Mem[11010]
011	N		
100	N		
101	N		
110	Y	10	Mem[10110]
111	N		

Cache Example

Word addr	Binary addr	Hit/miss	Cache block
16	10 000	Miss	000
3	00 011	Miss	011
16	10 000	Hit	000

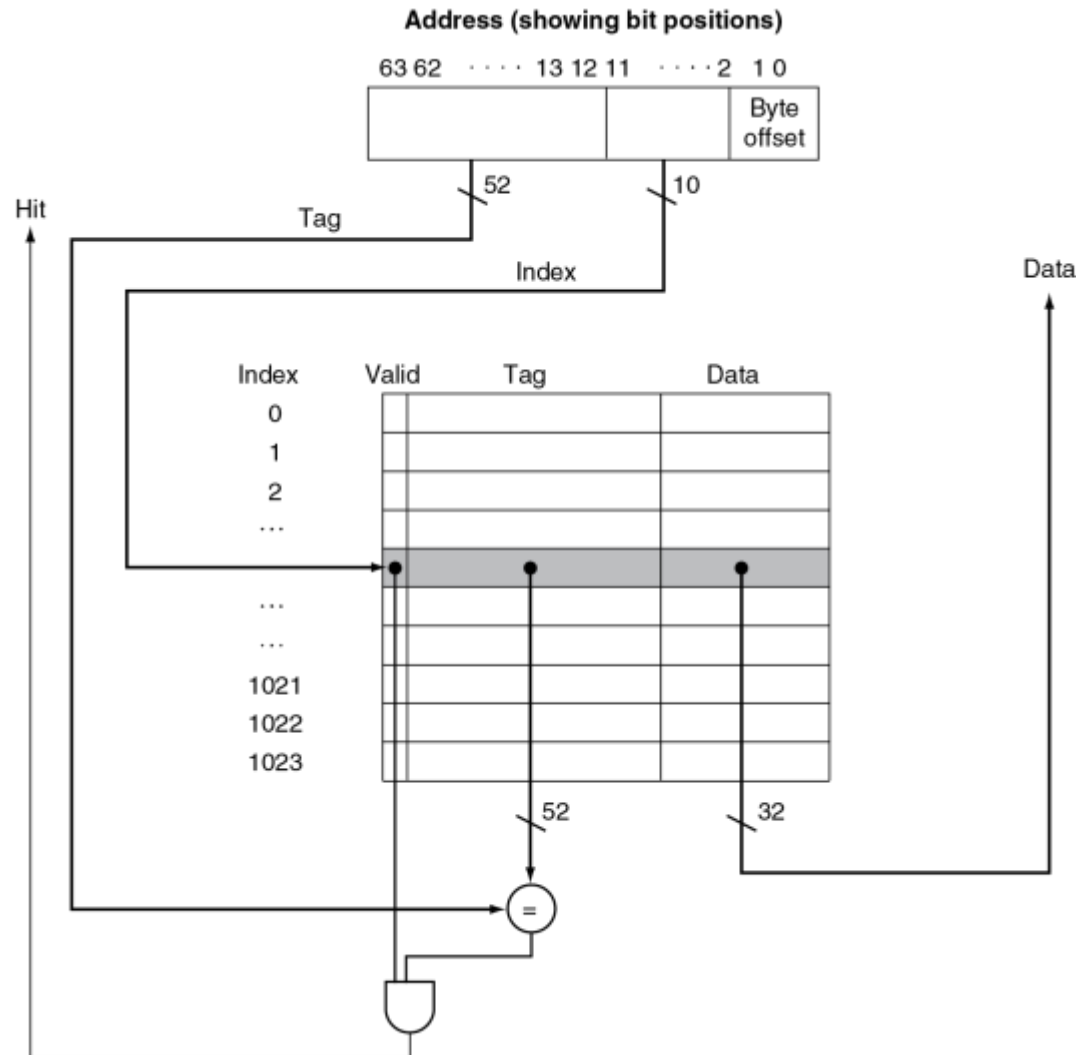
Index	V	Tag	Data
000	Y	10	Mem[10000]
001	N		
010	Y	11	Mem[11010]
011	Y	00	Mem[00011]
100	N		
101	N		
110	Y	10	Mem[10110]
111	N		

Cache Example

Word addr	Binary addr	Hit/miss	Cache block
18	10 010	Miss	010

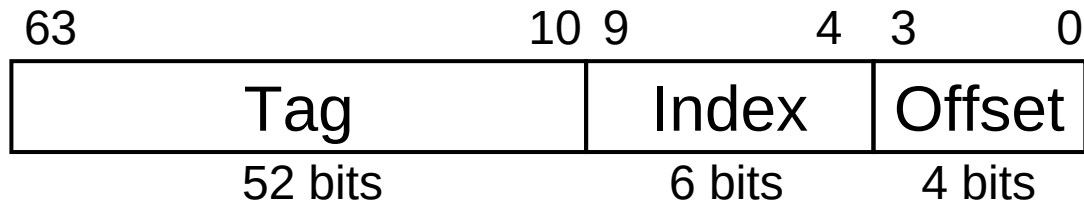
Index	V	Tag	Data
000	Y	10	Mem[10000]
001	N		
010	Y	10	Mem[10010]
011	Y	00	Mem[00011]
100	N		
101	N		
110	Y	10	Mem[10110]
111	N		

Address Subdivision



Example: Larger Block Size

- 64 blocks, 16 bytes/block
 - To what block number does address 1200 map?
- Block address = $\lfloor 1200/16 \rfloor = 75$
- Block number = $75 \text{ modulo } 64 = 11$



Block Size Considerations

- Larger blocks should reduce miss rate
 - Due to spatial locality
- But in a fixed-sized cache
 - Larger blocks $\Rightarrow$ fewer of them
 - More competition $\Rightarrow$ increased miss rate
 - Larger blocks $\Rightarrow$ pollution
- Larger miss penalty
 - Can override benefit of reduced miss rate
 - Early restart and critical-word-first can help

Cache Misses

- On cache hit, CPU proceeds normally
- On cache miss
 - Stall the CPU pipeline
 - Fetch block from next level of hierarchy
 - Instruction cache miss
 - Restart instruction fetch
 - Data cache miss
 - Complete data access

Write-Through

- On data-write hit, could just update the block in cache
 - But then cache and memory would be inconsistent
- Write through: also update memory
- But makes writes take longer
- Solution: write buffer
 - Holds data waiting to be written to memory
 - CPU continues immediately
 - Only stalls on write if write buffer is already full

Write-Back

- Alternative: On data-write hit, just update the block in cache
 - Keep track of whether each block is dirty
- When a dirty block is replaced
 - Write it back to memory
 - Can use a write buffer to allow replacing block to be read first

Write Allocation

- What should happen on a write miss?
- Alternatives for write-through
 - Allocate on miss: fetch the block
 - Write around: don't fetch the block
 - Since programs often write a whole block before reading it (e.g., initialization)
- For write-back
 - Usually fetch the block

Average Access Time

- Hit time is also important for performance
- Average memory access time (AMAT)
 - $\text{AMAT} = \text{Hit time} + \text{Miss rate} \times \text{Miss penalty}$
- Example
 - CPU with 1ns clock, hit time = 1 cycle, miss penalty = 20 cycles, l-cache miss rate = 5%
 - $\text{AMAT} = 1 + 0.05 \times 20 = 2\text{ns}$
 - 2 cycles per instruction

Performance Summary

- When CPU performance increased
 - Miss penalty becomes more significant
- Decreasing base CPI
 - Greater proportion of time spent on memory stalls
- Increasing clock rate
 - Memory stalls account for more CPU cycles
- Can't neglect cache behavior when evaluating system performance

Multilevel Caches

- Primary cache attached to CPU
 - Small, but fast
- Level-2 cache services misses from primary cache
 - Larger, slower, but still faster than main memory
- Main memory services L-2 cache misses
- Some high-end systems include L-3 cache

Multilevel Cache Considerations

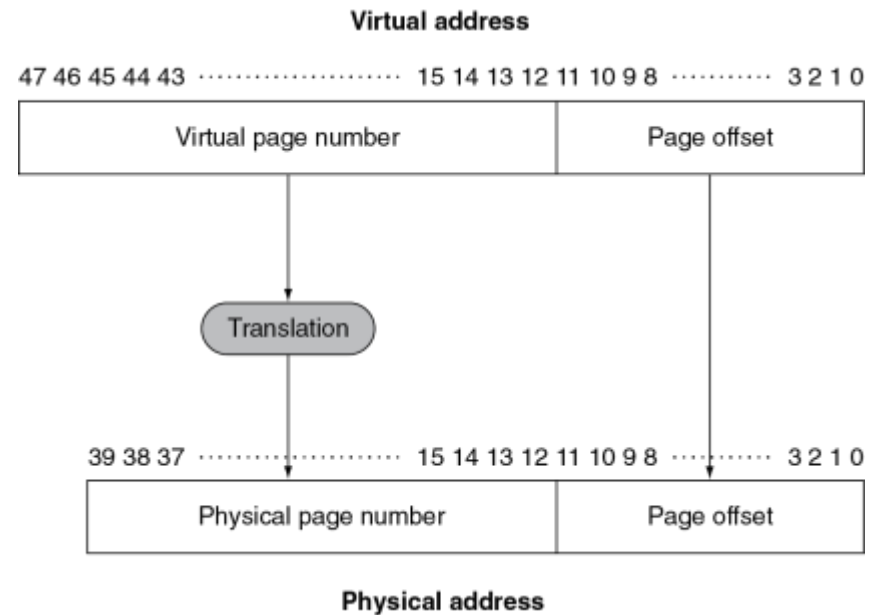
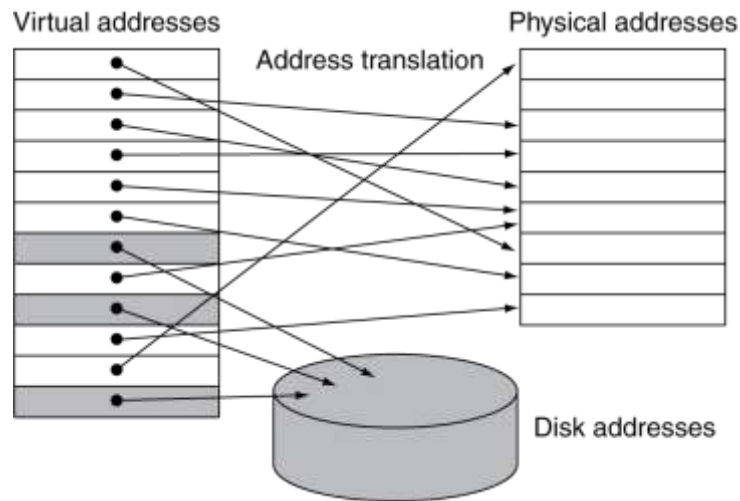
- Primary cache
 - Focus on minimal hit time
- L-2 cache
 - Focus on low miss rate to avoid main memory access
 - Hit time has less overall impact
- Results
 - L-1 cache usually smaller than a single cache
 - L-1 block size smaller than L-2 block size

Virtual Memory

- Use main memory as a “cache” for secondary (disk) storage
 - Managed jointly by CPU hardware and the operating system (OS)
- Programs share main memory
 - Each gets a private virtual address space holding its frequently used code and data
 - Protected from other programs
- CPU and OS translate virtual addresses to physical addresses
 - VM “block” is called a page
 - VM translation “miss” is called a page fault

Address Translation

- Fixed-size pages (e.g., 4K)



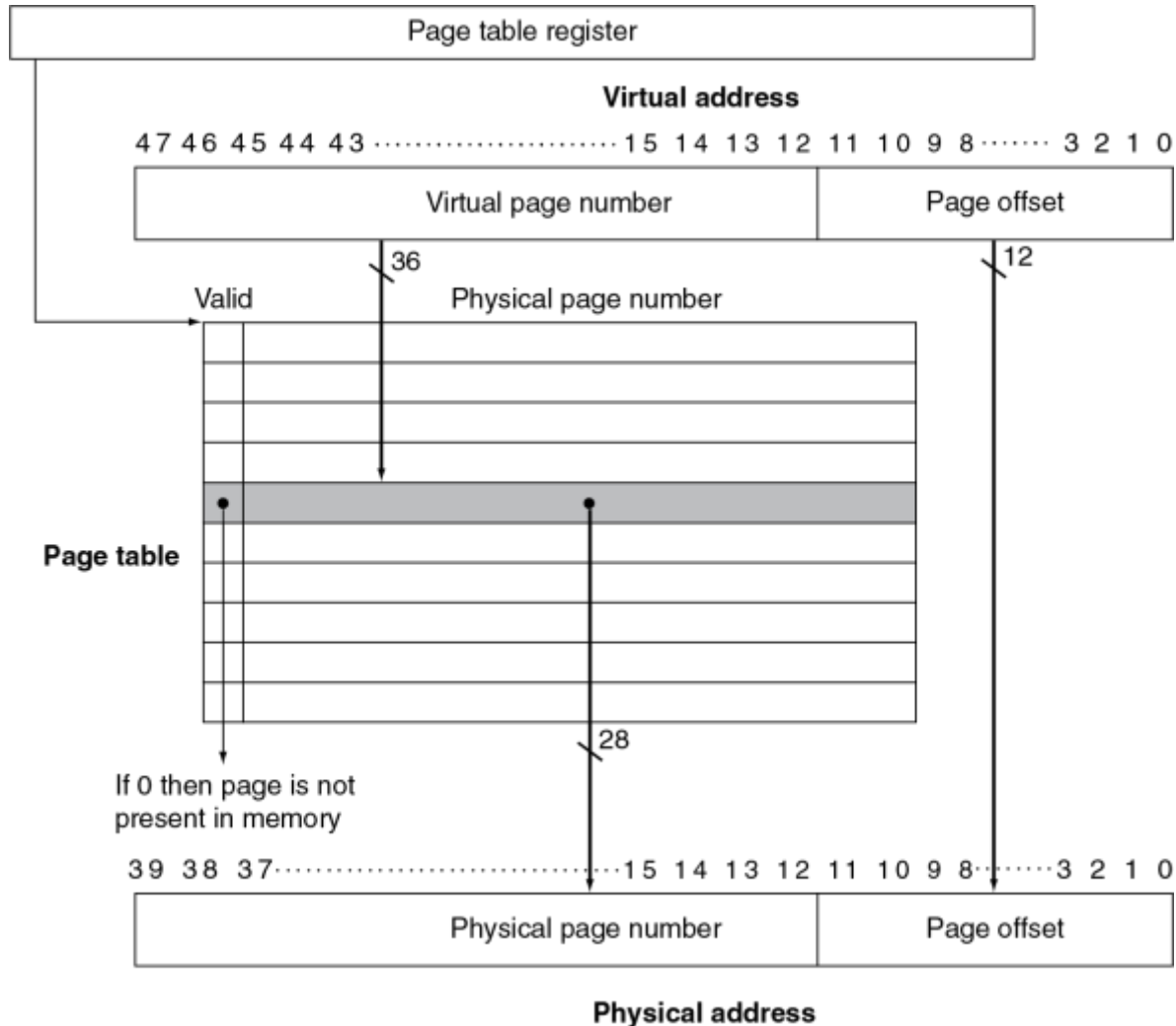
Page Fault Penalty

- On page fault, the page must be fetched from disk
 - Takes millions of clock cycles
 - Handled by OS code
- Try to minimize page fault rate
 - Fully associative placement
 - Smart replacement algorithms

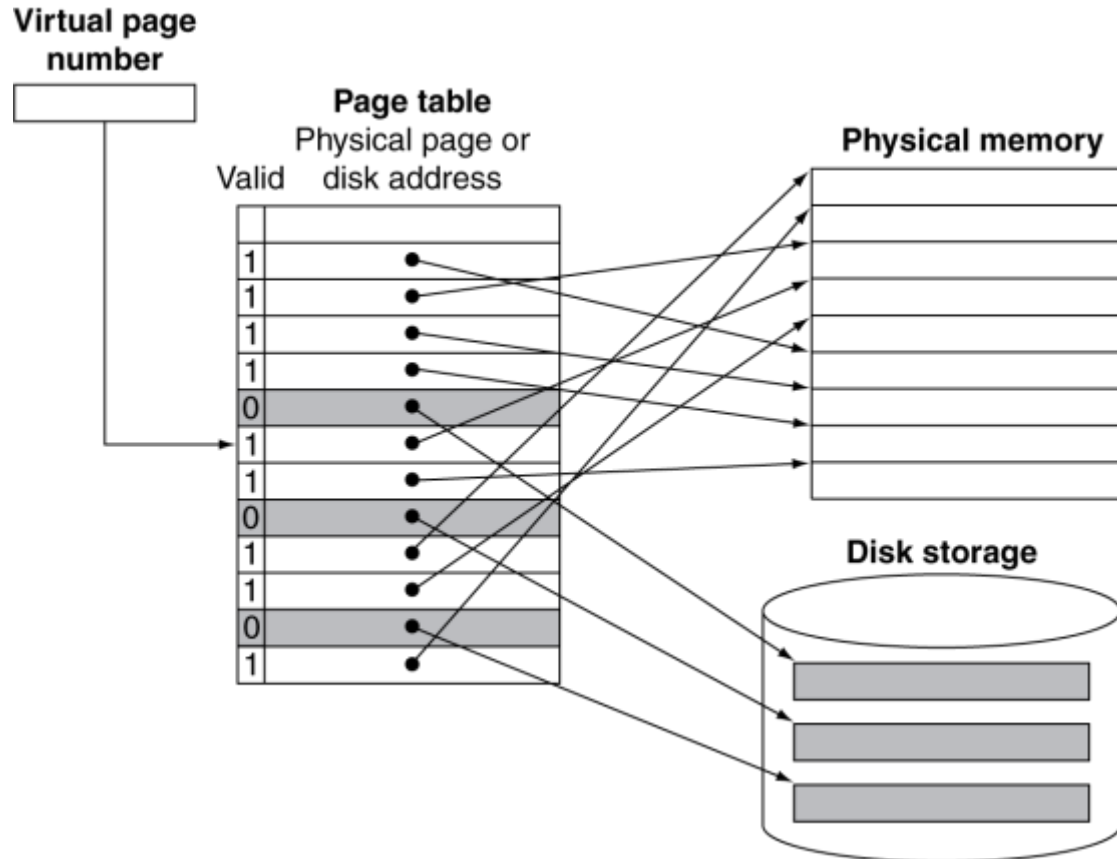
Page Tables

- Stores placement information
 - Array of page table entries, indexed by virtual page number
 - Page table register in CPU points to page table in physical memory
- If page is present in memory
 - PTE stores the physical page number
 - Plus other status bits (referenced, dirty, ...)
- If page is not present
 - PTE can refer to location in swap space on disk

Translation Using a Page Table



Mapping Pages to Storage



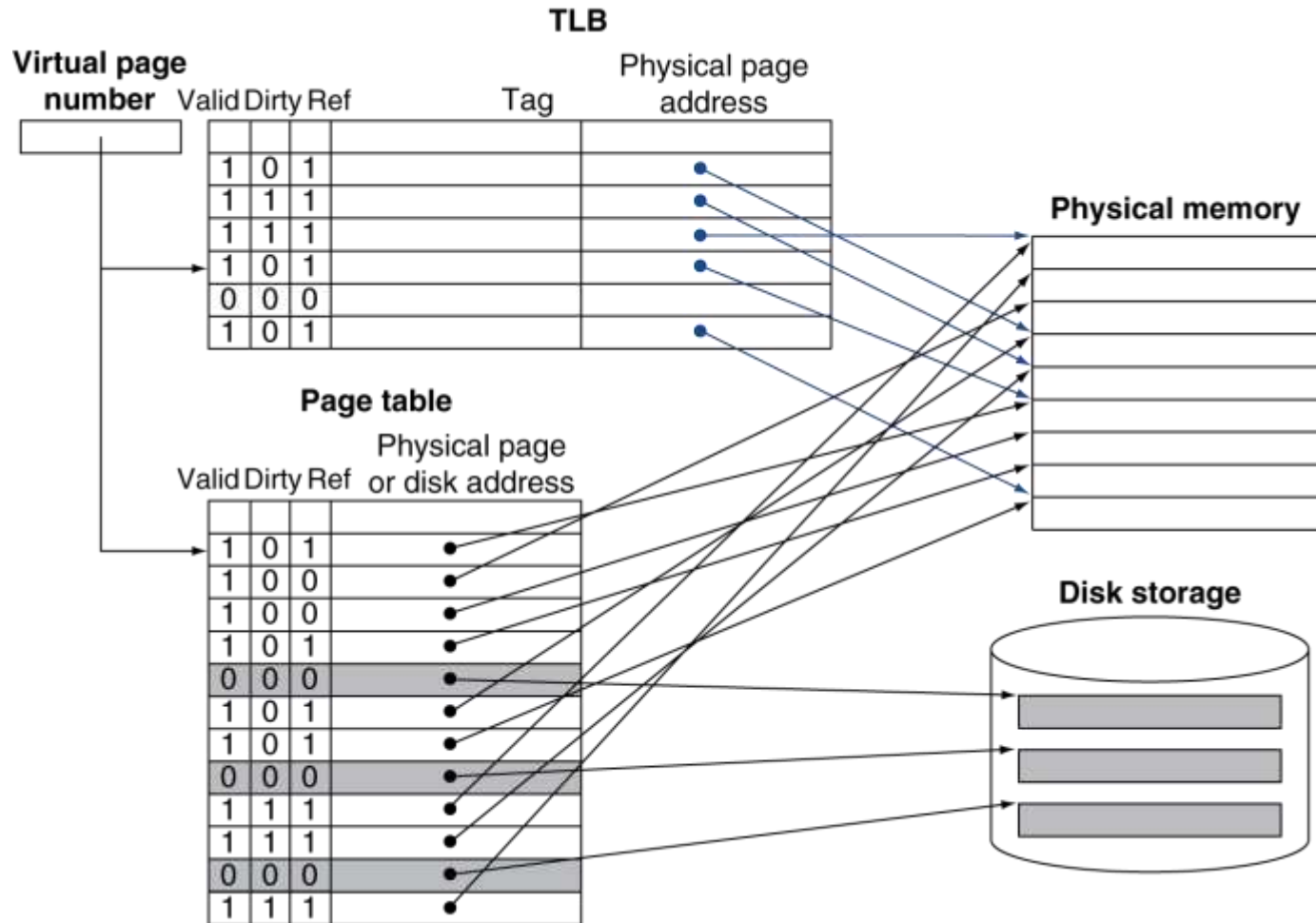
Replacement and Writes

- To reduce page fault rate, prefer least-recently used (LRU) replacement
 - Reference bit (aka use bit) in PTE set to 1 on access to page
 - Periodically cleared to 0 by OS
 - A page with reference bit = 0 has not been used recently
- Disk writes take millions of cycles
 - Block at once, not individual locations
 - Write through is impractical
 - Use write-back
 - Dirty bit in PTE set when page is written

Fast Translation Using a TLB

- Address translation would appear to require extra memory references
 - One to access the PTE
 - Then the actual memory access
- But access to page tables has good locality
 - So use a fast cache of PTEs within the CPU
 - Called a Translation Look-aside Buffer (TLB)
 - Typical: 16–512 PTEs, 0.5–1 cycle for hit, 10–100 cycles for miss, 0.01%–1% miss rate
 - Misses could be handled by hardware or software

Fast Translation Using a TLB



TLB Misses

- If page is in memory
 - Load the PTE from memory and retry
 - Could be handled in hardware
 - Can get complex for more complicated page table structures
 - Or in software
 - Raise a special exception, with optimized handler
- If page is not in memory (page fault)
 - OS handles fetching the page and updating the page table
 - Then restart the faulting instruction

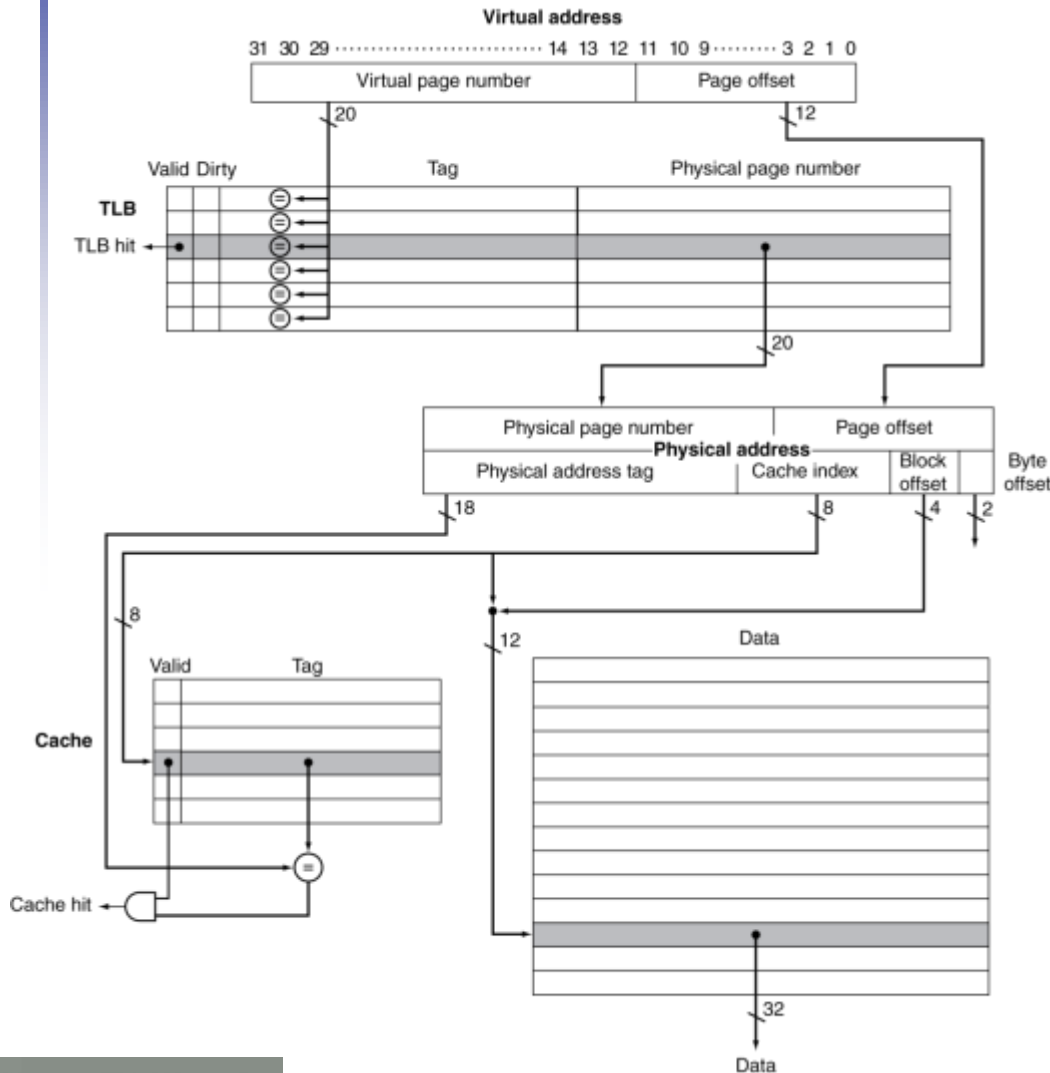
TLB Miss Handler

- TLB miss indicates
 - Page present, but PTE not in TLB
 - Page not present
- Must recognize TLB miss before destination register overwritten
 - Raise exception
- Handler copies PTE from memory to TLB
 - Then restarts instruction
 - If page not present, page fault will occur

Page Fault Handler

- Use faulting virtual address to find PTE
- Locate page on disk
- Choose page to replace
 - If dirty, write to disk first
- Read page into memory and update page table
- Make process runnable again
 - Restart from faulting instruction

TLB and Cache Interaction



- If cache tag uses physical address
 - Need to translate before cache lookup
- Alternative: use virtual address tag
 - Complications due to aliasing
 - Different virtual addresses for shared physical address

Memory Protection

- Different tasks can share parts of their virtual address spaces
 - But need to protect against errant access
 - Requires OS assistance
- Hardware support for OS protection
 - Privileged supervisor mode (aka kernel mode)
 - Privileged instructions
 - Page tables and other state information only accessible in supervisor mode
 - System call exception (e.g., ecall in RISC-V)

The Memory Hierarchy

The BIG Picture

- Common principles apply at all levels of the memory hierarchy
 - Based on notions of caching
- At each level in the hierarchy
 - Block placement
 - Finding a block
 - Replacement on a miss
 - Write policy